Dynamic Principal Component Analysis in Time Domain

Sangmyung Ha

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Theorem 1 (Spectral Isometry)**.** *Let $\{X_t\}_{t\in\mathbb{Z}}$ be a zero-mean, covariance-stationary $\mathbb{C}^n$-valued stochastic process with spectral distribution function $F(\lambda)$ and associated orthogonal-increment process $\{Z_X(\lambda)\}_{\lambda\in[-\pi,\pi]}$. Define a Hilbert space of square integrable functions*

$$\mathcal{L}^2(F) = \left\{ \varphi : [-\pi,\pi] \to \mathbb{C}^{n\times m} \;\middle|\; \int_{-\pi}^{\pi} \operatorname{tr}(\varphi(\lambda)\, dF(\lambda)\, \varphi(\lambda)^*) < \infty \right\},$$

with inner product defined by

$$\int_{-\pi}^{\pi} \operatorname{tr}(\varphi(\lambda)\, dF(\lambda)\, \psi(\lambda)^*)$$

for all $\varphi, \psi \in \mathcal{L}^2(F)$, where $*$ is complex conjugate transpose. Let $L^2(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{C}^m)$ be the usual Hilbert space of square integrable random vectors with inner product defined by

$$\mathbb{E}[X * Y]$$

for all $X, Y \in L^2(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{C}^m)$. Then, the mapping

$$\tau : \mathcal{L}^2(F) \to L^2(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{C}^m), \qquad \tau(\varphi) = \int_{-\pi}^{\pi} \varphi(\lambda)\, dZ_X(\lambda),$$

is a linear isometry. In particular, for all $\varphi, \psi \in \mathcal{L}^2(F)$,

$$\mathbb{E}\left[\left(\int \varphi(\lambda)\, dZ_X(\lambda) \right)^* \left(\int \psi(\lambda)\, dZ_X(\lambda) \right) \right] = \int_{-\pi}^{\pi} \operatorname{tr}(\varphi(\lambda)\, dF(\lambda)\, \psi(\lambda)^*)\,.$$

Proof. We first define the stochastic integral for step functions. Let φ be of the form

$$\varphi(\lambda) = \sum_{j=1}^{k} C_j \mathbf{1}_{(\lambda_{j-1},\lambda_j]}(\lambda),$$

where $-\pi = \lambda_0 < \lambda_1 < \cdots < \lambda_k = \pi$ and $C_j \in \mathbb{C}^{n\times m}$. Define

$$\int_{-\pi}^{\pi} \varphi(\lambda)\, dZ_X(\lambda) = \sum_{j=1}^{k} C_j \big(Z_X(\lambda_j) - Z_X(\lambda_{j-1}) \big).$$

Since $Z_X(\lambda)$ has orthogonal increments,

$$\mathbb{E}\Big[\Big(\int \varphi(\lambda)\,dZ_X(\lambda)\Big)^* \Big(\int \psi(\lambda)\,dZ_X(\lambda)\Big)\Big] = \sum_{j=1}^{k}\sum_{\ell=1}^{k} \mathrm{tr}\big(C_j\,\mathbb{E}\big[(Z_X(\lambda_j)-Z_X(\lambda_{j-1}))(Z_X(\mu_\ell)-Z_X(\mu_{\ell-1}))^*\big]D_\ell^*\big)$$
$$= \sum_{j=1}^{k} \mathrm{tr}\big(C_j\big(F(\lambda_j)-F(\lambda_{j-1})\big)D_j^*\big)\,,$$

where $\psi(\lambda)=\sum_{\ell=1}^{k} D_\ell \mathbf{1}_{(\mu_{\ell-1},\mu_\ell]}(\lambda)$. This equals

$$\int_{-\pi}^{\pi} \mathrm{tr}(\varphi(\lambda)\,dF(\lambda)\,\psi(\lambda)^*)\,.$$

Thus the isometry holds for step functions. Since step functions are dense in $\mathcal{L}^2(F)$ and τ is continuous under the induced norm, the identity extends to all $\varphi,\psi\in\mathcal{L}^2(F)$ by completion. $\square$

1 Dynamic PCA in the Frequency Domain

Brillinger (1981) introduces dynamic principal components as a dimension-reduction method based on an optimal reconstruction criterion.

Let $\{X_t\}_{t\in\mathbb{Z}}$ be an n-dimensional, zero-mean, weakly stationary time series with absolutely summable autocovariance matrices

$$\Gamma_X(k)=\mathbb{E}[X_{t+k}X_t^*]\,,\qquad k\in\mathbb{Z},$$

where * denotes the conjugate transpose. The associated $n\times n$ spectral density matrix is given by

$$f_X(\lambda)=\sum_{k=-\infty}^{\infty}\Gamma_X(k)e^{i\lambda k},\qquad \lambda\in[-\pi,\pi].$$

The goal of dimension reduction is to construct a q-dimensional filtered process, with $q<n$, that captures as much of the variation in X_t as possible. To this end, we seek a sequence of filter coefficients $\{b_k\}_{k\in\mathbb{Z}}$, where each b_k is a $q\times n$ matrix, such that the filtered process

$$\xi_t=\sum_{k=-\infty}^{\infty} b_kX_{t-k}=b(L)X_t$$

provides a low-dimensional representation of X_t. Here, $b(L)=\sum_{k=-\infty}^{\infty}b_kL^k$ denotes the corresponding matrix-valued lag polynomial.

To assess the quality of this representation, we also introduce an $n\times q$ matrix-valued filter $c(L)=\sum_{k=-\infty}^{\infty}c_kL^k$, which is used to reconstruct X_t from ξ_t via

$$\widetilde{X}_t=\sum_{k=-\infty}^{\infty}c_k\xi_{t-k}=c(L)\xi_t.$$

The filters $b(L)$ and $c(L)$ are chosen jointly to minimize the mean squared reconstruction error

$$\mathbb{E}\Big[(X_t - \widetilde{X}_t)^*(X_t - \widetilde{X}_t)\Big] = \mathbb{E}[(X_t - c(L)b(L)X_t)^*(X_t - c(L)b(L)X_t)]\,.$$

This optimization problem leads to the dynamic principal components of X_t.

Theorem 2 (Brillinger (1981), Theorem 9.3.1)**.** *Let $\{X_t\}$ be an n-dimensional, zero-mean, covariance-stationary process with spectral density matrix $f_X(\lambda)$. Among all rank-q linear filters of the form*

$$\widetilde{X}_t = c(L)b(L)X_t,$$

where $b(L)$ is $q \times n$ and $c(L)$ is $n \times q$, the filters $\{b_k\}$ and $\{c_k\}$ that jointly minimize

$$\mathbb{E}\Big[(X_t - \widetilde{X}_t)^*(X_t - \widetilde{X}_t)\Big] = \mathbb{E}[(X_t - c(L)b(L)X_t)^*(X_t - c(L)b(L)X_t)]$$

are given by

$$b_k = \frac{1}{2\pi}\int_{-\pi}^{\pi} b(e^{i\lambda})e^{-ik\lambda}\,d\lambda, \qquad c_k = \frac{1}{2\pi}\int_{-\pi}^{\pi} c(e^{i\lambda})e^{-ik\lambda}\,d\lambda,$$

where

$$b(e^{i\lambda}) = \begin{pmatrix} u_1(\lambda) & \cdots & u_q(\lambda)\end{pmatrix}^*, \qquad c(e^{i\lambda}) = b(e^{i\lambda})^*,$$

and $\{u_j(\lambda)\}_{j=1}^q$ are the eigenvectors of $f_X(\lambda)$ corresponding to its q largest eigenvalues.

Proof. Define the reconstructed process

$$\widetilde{X}_t = A(L)X_t, \qquad A(L) = c(L)b(L),$$

where $A(L)$ is an $n\times n$ linear filter of rank at most q. The goal is to choose $A(L)$ to minimize the mean squared reconstruction error. By the spectral representation theorem,

$$X_t = \int_{-\pi}^{\pi} e^{-it\lambda}\,dZ_X(\lambda),$$

where $Z_X(\lambda)$ is an orthogonal increment process with

$$\mathbb{E}[Z_X(\lambda)Z_X(\lambda)^*] = F_X(\lambda),$$

where $F_X(\lambda)$ is the spectral distribution of $\{X_t\}$. The filtered process $\widetilde{X}_t$ admits the representation

$$\widetilde{X}_t = \int_{-\pi}^{\pi} e^{-it\lambda}A(e^{i\lambda})\,dZ_X(\lambda).$$

Hence,

$$X_t - \widetilde{X}_t = \int_{-\pi}^{\pi} e^{-it\lambda}\big(I - A(e^{i\lambda})\big)\,dZ_X(\lambda).$$

Using the isometry of the spectral representation in Theorem 1,

$$\mathbb{E}\Big[(X_t - \widetilde{X}_t)^*(X_t - \widetilde{X}_t)\Big] = \operatorname{tr}\bigg(\int_{-\pi}^{\pi}\big(I - A(e^{i\lambda})\big)f_X(\lambda)\big(I - A(e^{i\lambda})\big)^*\frac{d\lambda}{2\pi}\bigg)\,.$$

Since the integrand is nonnegative definite, minimization of the integral reduces to pointwise minimization in λ. Write

$$f_X(\lambda) = f_X(\lambda)^{1/2} f_X(\lambda)^{1/2}.$$

Then

$$\big(I - A\big) f_X \big(I - A\big)^* = \big(f_X^{1/2} - A f_X^{1/2}\big)\big(f_X^{1/2} - A f_X^{1/2}\big)^*.$$

Thus, for each fixed λ, the problem reduces to approximating $f_X(\lambda)^{1/2}$ by $A(e^{i\lambda}) f_X(\lambda)^{1/2}$ in Frobenius norm, subject to $\operatorname{rank}(A) \leq q$. By the Eckart–Young–Mirsky theorem, the minimizer is the orthogonal projection onto the subspace spanned by the q leading eigenvectors of $f_X(\lambda)$. Therefore,

$$A(e^{i\lambda}) = \sum_{j=1}^{q} u_j(\lambda) u_j(\lambda)^*,$$

where $u_j(\lambda)$ are the eigenvectors associated with the q largest eigenvalues of $f_X(\lambda)$. Writing $A(e^{i\lambda}) = c(e^{i\lambda}) b(e^{i\lambda})$ with

$$b(e^{i\lambda}) = \begin{pmatrix} u_1(\lambda) & \cdots & u_q(\lambda) \end{pmatrix}^*, \qquad c(e^{i\lambda}) = b(e^{i\lambda})^*,$$

yields the desired factorization.

Finally, since $b(e^{i\lambda})$ and $c(e^{i\lambda})$ are square-integrable matrix-valued functions on $[-\pi, \pi]$, they admit Fourier series representations converging in L^2:

$$b(e^{i\lambda}) = \sum_{k=-\infty}^{\infty} b_k e^{ik\lambda}, \qquad c(e^{i\lambda}) = \sum_{k=-\infty}^{\infty} c_k e^{ik\lambda},$$

with coefficients

$$b_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} b(e^{i\lambda}) e^{-ik\lambda}\, d\lambda, \qquad c_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} c(e^{i\lambda}) e^{-ik\lambda}\, d\lambda.$$

These coefficients define the optimal linear filters $b(L)$ and $c(L)$. □

Note that the reconstruction filter satisfies $c_k = b'_{-k}$. To see this, recall that X_t is real-valued, which implies that its spectral density matrix satisfies the Hermitian symmetry

$$\overline{f_X(\lambda)} = \sum_{k=-\infty}^{\infty} \Gamma_X(k) e^{-i\lambda k} = f_X(-\lambda).$$

Consequently, the frequency response of the filter $b(L)$ satisfies

$$\overline{b(e^{i\lambda})} = b(e^{-i\lambda}).$$

Since the optimal reconstruction filter is given by

$$c(e^{i\lambda}) = b(e^{i\lambda})^*,$$

it follows that

$$c(e^{i\lambda}) = b(e^{-i\lambda})'.$$

Taking inverse Fourier transforms, we obtain

$$\begin{aligned} c_k &= \frac{1}{2\pi}\int_{-\pi}^{\pi} c(e^{i\lambda})e^{-ik\lambda}\,d\lambda \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} b(e^{-i\lambda})'e^{-ik\lambda}\,d\lambda \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} b(e^{i\lambda})'e^{-i(-k)\lambda}\,d\lambda \\ &= b'_{-k}, \end{aligned}$$

where the third equality follows from the change of variables $\lambda \mapsto -\lambda$.

Let a low-dimensional approximate representation of $\{X_t\}$ be defined by

$$\xi_t = b(L)X_t,$$

which we refer to as the *dynamic principal components* of $\{X_t\}$. We now derive several key properties of ξ_t.

First, the spectral density matrix of $\xi_t = b(L)X_t$ is

$$f_\xi(\lambda) = b(e^{i\lambda})f_X(\lambda)b(e^{i\lambda})^*,$$

which is a $q \times q$ diagonal matrix of the form

$$f_\xi(\lambda) = \begin{pmatrix} \mu_1(\lambda) & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \mu_q(\lambda) \end{pmatrix}, \qquad \lambda \in [-\pi, \pi].$$

Thus, the components of ξ_t are mutually orthogonal at all frequencies: for $i \neq j$, the processes ξ_{it} and ξ_{jt} have zero coherence at every frequency.

This frequency-domain orthogonality implies time-domain orthogonality at all leads and lags. Indeed, the autocovariance matrix of ξ_t,

$$\Gamma_\xi(k) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f_\xi(\lambda)e^{-ik\lambda}\,d\lambda,$$

is diagonal for every k. Explicitly,

$$\Gamma_\xi(k) = \frac{1}{2\pi}\begin{pmatrix} \int_{-\pi}^{\pi}\mu_1(\lambda)e^{-ik\lambda}\,d\lambda & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \int_{-\pi}^{\pi}\mu_q(\lambda)e^{-ik\lambda}\,d\lambda \end{pmatrix}.$$

Consequently, for $i \neq j$, ξ_{it} and ξ_{jt} are uncorrelated at all leads and lags. This property is stronger than the orthogonality delivered by conventional (static) PCA, which guarantees only contemporaneous uncorrelatedness.

In particular, when $k = 0$,

$$\Gamma_\xi(0) = \frac{1}{2\pi}\begin{pmatrix} \int_{-\pi}^{\pi}\mu_1(\lambda)\,d\lambda & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \int_{-\pi}^{\pi}\mu_q(\lambda)\,d\lambda \end{pmatrix}.$$

At each frequency λ, the quantities $\mu_j(\lambda)$, $j = 1, \dots, q$, are the maximal values attainable under the normalization constraint

$$b(e^{i\lambda})b(e^{i\lambda})^* = I_q.$$

Therefore, the matrix-valued filter $b(L)$ maximizes the trace of the variance matrix $\Gamma_\xi(0)$ subject to this constraint. As shown below, this variational characterization is a defining feature of dynamic PCA in the frequency domain.

It is important to note, however, that the definition of dynamic PCA does not directly impose optimality on $\Gamma_\xi(k)$ for $k \neq 0$. In general, the traces of the autocovariance matrices $\Gamma_\xi(k)$, $k \neq 0$, need not be maximized by the dynamic PCA solution.

Claim. *Minimizing the mean squared reconstruction error in Theorem 2 is equivalent to maximizing the variance of*

$$\xi_t = b(L)X_t$$

subject to the constraints

$$\sum_{k=-\infty}^{\infty} b_k b_k' = I_q, \tag{1}$$

$$\sum_{p=-\infty}^{\infty} b_p b_{p+j}' = 0, \qquad j = 1, 2, \dots, \tag{2}$$

referred to as the static *and* dynamic *constraints, respectively. Moreover, at the optimum the reconstruction filter satisfies*

$$c_k = b_{-k}'.$$

Proof. Let $\widetilde{X}_t = c(L)b(L)X_t$ and define $\xi_t = b(L)X_t$. We show that, under the normalization

$$b(e^{i\lambda})b(e^{i\lambda})^* = I_q \quad \text{and} \quad c(e^{i\lambda}) = b(e^{i\lambda})^*,$$

the sum of the mean squared reconstruction error and the variance of ξ_t equals the variance of X_t. Equivalently,

$$\mathbb{E}\Big[(X_t - \widetilde{X}_t)^*(X_t - \widetilde{X}_t)\Big] = \operatorname{tr}(\Gamma_X(0)) - \operatorname{tr}(\Gamma_\xi(0))\,. \tag{3}$$

This decomposition implies that minimizing the reconstruction error is equivalent to maximizing the variance of the dynamic principal components. We then show that the frequency-domain condition

$$b(e^{i\lambda})b(e^{i\lambda})^* = I_q$$

is equivalent to the time-domain restrictions (1) and (2). Note that the condition $c(e^{i\lambda}) = b(e^{i\lambda})^*$ holds at the solution of the reconstruction problem and does not enter directly into the variance-maximization formulation. Expanding the quadratic form,

$$\mathbb{E}[(X_t - \widetilde{X}_t)^*(X_t - \widetilde{X}_t)] = \mathbb{E}[X_t^* X_t] - \mathbb{E}[\widetilde{X}_t^* X_t] - \mathbb{E}[X_t^* \widetilde{X}_t] + \mathbb{E}[\widetilde{X}_t^* \widetilde{X}_t].$$

Clearly,

$$\mathbb{E}[X_t^* X_t] = \operatorname{tr}(\Gamma_X(0)).$$

Using the spectral representations

$$X_t = \int_{-\pi}^{\pi} e^{-it\lambda}\, dZ_X(\lambda), \qquad \widetilde{X}_t = \int_{-\pi}^{\pi} e^{-it\lambda} c(e^{i\lambda}) b(e^{i\lambda})\, dZ_X(\lambda),$$

and the spectral isometry, we obtain

$$\mathbb{E}[\widetilde{X}_t^* X_t] = \operatorname{tr}\Big(\int_{-\pi}^{\pi} b(e^{i\lambda}) f_X(\lambda) b(e^{i\lambda})^* \frac{d\lambda}{2\pi}\Big) = \operatorname{tr}\big(\Gamma_\xi(0)\big).$$

Similarly,

$$\mathbb{E}[\widetilde{X}_t^* \widetilde{X}_t] = \operatorname{tr}\Big(\int_{-\pi}^{\pi} b(e^{i\lambda}) f_X(\lambda) b(e^{i\lambda})^* \frac{d\lambda}{2\pi}\Big) = \operatorname{tr}\big(\Gamma_\xi(0)\big),$$

where we used $b(e^{i\lambda})b(e^{i\lambda})^* = I_q$. Substituting these expressions yields (3).

We now show that the frequency-domain constraint

$$b(e^{i\lambda})b(e^{i\lambda})^* = I_q \quad \text{for all } \lambda$$

is equivalent to the time-domain constraints (1)–(2). Write

$$b(e^{i\lambda}) = \sum_{k=-\infty}^{\infty} b_k e^{ik\lambda}.$$

Then

$$b(e^{i\lambda})b(e^{i\lambda})^* = \sum_{j=-\infty}^{\infty} \left(\sum_{p=-\infty}^{\infty} b_p b'_{p+j} \right) e^{ij\lambda},$$

under general conditions that allow us to use Fubini's theorem for infinite series. Since $\{e^{ij\lambda}\}_{j\in\mathbb{Z}}$ form an orthogonal basis of $L^2([-\pi,\pi])$, the identity $b(e^{i\lambda})b(e^{i\lambda})^* = I_q$ for all λ holds if and only if

$$\sum_{p=-\infty}^{\infty} b_p b'_p = I_q, \qquad \sum_{p=-\infty}^{\infty} b_p b'_{p+j} = 0 \text{ for } j \neq 0,$$

which are precisely (1) and (2).

Finally, given $b(e^{i\lambda})b(e^{i\lambda})^* = I_q$, the choice $c(e^{i\lambda}) = b(e^{i\lambda})^*$ minimizes the mean squared error pointwise in frequency. Taking Fourier coefficients yields $c_k = b'_{-k}$. □

2 Dynamic PCA in the Time Domain

In practice, we cannot maximize the variance of $\xi_t = \sum_{k=-\infty}^{\infty} b_k X_{t-k}$ over infinite sequences $\{b_k\}_{k=-\infty}^{\infty}$. For estimation, we need to use finite-length filters $b(L) = \sum_{k=-m}^{m} b_k L^k$. Now, we show some results using finite-length filters. Suppose now that the dynamic principal components are written as

$$\xi_t = \sum_{k=-m}^{m} b_k X_{t-k}$$

Furthermore, we seek to recover X_t by

$$\widetilde{X}_t = \sum_{k=-m}^{m} c_k \xi_{t-k}$$

As before, we assume that $\{b_k\}_{k=-m}^{m}$ and $\{c_k\}_{k=-m}^{m}$ are $q \times n$ and $n \times q$ dimensional matrices respectively. If the analog of static and dynamic constraints hold in finite length filters, the following result holds.

Claim. *Given that the coefficients satisfy the property*

$$c_k = b'_{-k}, \quad \forall k = -m, ..., m \tag{4}$$

$$\sum_{k=-m}^{m} b_k b'_k = I_q \tag{5}$$

$$\sum_{p=-m}^{m-j} b_p b'_{p+j} = 0, \quad \forall j = 1, ..., 2m \tag{6}$$

we have

$$\mathbb{E}\left[(X_t - \widetilde{X}_t)^*(X_t - \widetilde{X}_t)\right] = tr(\Gamma_X(0)) - tr(\Gamma_\xi(0))$$

Thus, maximizing the variance of ξ_t under the constraint (4), (5), *and* (6) *is equivalent to minimizing the mean squared error under the same constraints.*

Proof. Note that

$$\mathbb{E}\left[(X_t - \widetilde{X}_t)^*(X_t - \widetilde{X}_t)\right] = \mathbb{E}\left[X_t^* X_t\right] - \mathbb{E}\left[\widetilde{X}_t^* X_t\right] - \mathbb{E}\left[X_t^* \widetilde{X}_t\right] + \mathbb{E}\left[\widetilde{X}_t^* \widetilde{X}_t\right]$$

We show that $\mathbb{E}\left[\widetilde{X}_t^* X_t\right] = \mathbb{E}\left[X_t^* \widetilde{X}_t\right] = \mathbb{E}\left[\widetilde{X}_t^* \widetilde{X}_t\right] = \mathrm{tr}\left(\Gamma_\xi(0)\right)$ under the constraints (4), (5), and (6). First, note that the autocovariance of ξ_t is given by

$$\begin{aligned}
\Gamma_\xi(h) &= \mathbb{E}[\xi_{t+h}\xi_t^*] \\
&= \mathbb{E}\left[\left(\sum_{k=-m}^{m} b_k X_{t+h-k}\right)\left(\sum_{j=-m}^{m} b_j X_{t-j}\right)^*\right] \\
&= \sum_{k=-m}^{m}\sum_{j=-m}^{m} b_k \Gamma_X(h+j-k) b'_j
\end{aligned}$$

Thus, we have

$$
\begin{aligned}
\mathbb{E}\left[\widetilde{X}_t^* X_t\right] &= \mathbb{E}\left[\left(\sum_{k=-m}^{m} c_k \xi_{t-k}\right)^* X_t\right] \\
&= \operatorname{tr}\left(\mathbb{E}\left[X_t \left(\sum_{k=-m}^{m} c_k \xi_{t-k}\right)^*\right]\right) \\
&= \operatorname{tr}\left(\sum_{k=-m}^{m} \mathbb{E}\left[X_t \xi_{t-k}^*\right] c_k'\right) \\
&= \operatorname{tr}\left(\sum_{k=-m}^{m} \mathbb{E}\left[X_t \left(\sum_{j=-m}^{m} b_j X_{t-k-j}\right)^*\right] c_k'\right) \\
&= \operatorname{tr}\left(\sum_{k=-m}^{m} \sum_{j=-m}^{m} \mathbb{E}\left[X_t X_{t-k-j}^*\right] b_j' c_k'\right) \\
&= \operatorname{tr}\left(\sum_{k=-m}^{m} \sum_{j=-m}^{m} \Gamma_X(k+j) b_j' c_k'\right) \\
&= \operatorname{tr}\left(\sum_{k=-m}^{m} \sum_{j=-m}^{m} c_k' \Gamma_X(j+k) b_j'\right) \\
&= \operatorname{tr}\left(\sum_{k=-m}^{m} \sum_{j=-m}^{m} b_k \Gamma_X(j-k) b_j'\right) \\
&= \operatorname{tr}\left(\Gamma_\xi(0)\right)
\end{aligned}
$$

where we've used the condition $c_k = b_{-k}'$. $\mathbb{E}\left[X_t^* \widetilde{X}_t\right] = \operatorname{tr}\left(\Gamma_\xi(0)\right)$ can be shown similarly.

Finally,

$$
\begin{aligned}
\mathbb{E}\left[\widetilde{X}_t^* \widetilde{X}_t\right] &= \operatorname{tr}\left(\mathbb{E}\left[\widetilde{X}_t \widetilde{X}_t^*\right]\right) \\
&= \operatorname{tr}\left(\mathbb{E}\left[\left(\sum_{k=-m}^{m} c_k \xi_{t-k}\right)\left(\sum_{j=-m}^{m} c_j \xi_{t-j}\right)^*\right]\right) \\
&= \operatorname{tr}\left(\sum_{k=-m}^{m}\sum_{j=-m}^{m} c_k\, \mathbb{E}\left[\xi_{t-k}\xi_{t-j}^*\right] c_j'\right) \\
&= \operatorname{tr}\left(\sum_{k=-m}^{m}\sum_{j=-m}^{m} c_k \Gamma_\xi(j-k) c_j'\right) \\
&= \operatorname{tr}\left(\sum_{k=-m}^{m}\sum_{j=-m}^{m} c_k \left(\sum_{p=-m}^{m}\sum_{q=-m}^{m} b_p \Gamma_X(j-k+q-p) b_q'\right) c_j'\right) \\
&= \operatorname{tr}\left(\sum_{k=-m}^{m}\sum_{j=-m}^{m}\sum_{p=-m}^{m}\sum_{q=-m}^{m} c_k b_p \Gamma_X(j-k+q-p) b_q' c_j'\right) \\
&= \operatorname{tr}\left(\sum_{k=-m}^{m}\sum_{j=-m}^{m}\sum_{p=-m}^{m}\sum_{q=-m}^{m} b_k' b_p \Gamma_X(k-j+q-p) b_q' b_j\right)
\end{aligned}
$$

where we've used the constraint $c_k = b_{-k}'$ in the last equality. Now, we use the condition $\sum_{k=-m}^{m} b_k b_k' = I_q$, and $\sum_{p=-m}^{m-j} b_p b_{p+j}' = 0$, $\forall j = 1, ..., 2m$. We see that

$$
\begin{aligned}
\mathbb{E}\left[\widetilde{X}_t^* \widetilde{X}_t\right] &= \operatorname{tr}\left(\sum_{k=-m}^{m}\sum_{j=-m}^{m}\sum_{p=-m}^{m}\sum_{q=-m}^{m} b_k' b_p \Gamma_X(k-j+q-p) b_q' b_j\right) \\
&= \sum_{p=-m}^{m}\sum_{q=-m}^{m}\sum_{k=-m}^{m}\sum_{j=-m}^{m} \operatorname{tr}\left(b_q' b_j b_k' b_p \Gamma_X(k-j+q-p)\right) \\
&= \sum_{p=-m}^{m}\sum_{q=-m}^{m}\left(\sum_{l=-m}^{m} \operatorname{tr}\left(b_q' b_l b_l' b_p \Gamma_X(q-p)\right)\right. \\
&\qquad\left. + \sum_{j=1}^{2m}\sum_{l=-m}^{m-j}\left[\operatorname{tr}\left(b_q' b_l b_{l+j}' b_p \Gamma_X(q-p+j)\right) + \operatorname{tr}\left(b_q' b_{l+j} b_l' b_p \Gamma_X(q-p-j)\right)\right]\right)
\end{aligned}
$$

Here, we've reordered the summations in terms of $k = j$, and $k \neq j$. Since $\sum_{l=-m}^{m} b_l b_l' = I_q$, and $\sum_{l=-m}^{m} b_{l+j} b_l' = 0$ for all $j = 1, \ldots, 2m$, we have

$$\begin{aligned}
\mathbb{E}\left[\widetilde{X}_t^* \widetilde{X}_t\right] &= \sum_{p=-m}^{m} \sum_{q=-m}^{m} \left(\sum_{l=-m}^{m} \operatorname{tr}\left(b_q' b_l b_l' b_p \Gamma_X(q-p)\right) \right. \\
&\qquad \left. + \sum_{j=1}^{2m} \sum_{l=-m}^{m-j} \left[\operatorname{tr}\left(b_q' b_l b_{l+j}' b_p \Gamma_X(q-p+j)\right) + \operatorname{tr}\left(b_q' b_{l+j} b_l' b_p \Gamma_X(q-p-j)\right)\right] \right) \\
&= \sum_{p=-m}^{m} \sum_{q=-m}^{m} \operatorname{tr}\left(b_q' b_p \Gamma_X(q-p)\right) \\
&= \sum_{p=-m}^{m} \sum_{q=-m}^{m} \operatorname{tr}\left(b_p \Gamma_X(q-p) b_q'\right) \\
&= \operatorname{tr}\left(\Gamma_\xi(0)\right)
\end{aligned}$$

Thus, we have

$$\mathbb{E}\left[(X_t - \widetilde{X}_t)^*(X_t - \widetilde{X}_t)\right] = \operatorname{tr}\left(\Gamma_X(0)\right) - \operatorname{tr}\left(\Gamma_\xi(0)\right)$$

□